Envelope-Domain Preprocessing for Ultra-Compact LSTM-Based Broken Rotor Bar Severity Grading

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AI-assisted tools were used for language refinement and clarity. All technical content, experiments, and conclusions are the authors' own.

Abstract

We present a 4,773-parameter LSTM classifier for 5-class broken rotor bar severity grading from three-phase stator currents. On the public Universidade Federal de Uberlândia dataset (IEEE DataPort), it reaches 88.0% accuracy and trains in under 3 minutes on a laptop CPU, entirely in a web browser.

A preprocessing pipeline extracts the amplitude envelope of the stator current via half-cycle moving RMS, decimates it to 60 Hz, and removes the load-dependent baseline by per-window mean subtraction. This collapses the input from 256 high-frequency samples to 64 slowly-varying envelope samples. Without it, the same LSTM fails to train due to vanishing gradients through the longer sequence.

Broken rotor bars produce sidebands at $f \pm 2sf$ in the stator current spectrum. In the time domain these appear as slow amplitude modulation at 2–6 Hz, which the moving RMS captures directly. The envelope captures the discriminative information relevant to BRB severity, while the carrier primarily contributes redundant high-frequency structure. An ablation across four pipeline variants (26.8% to 88.0%) confirms that each preprocessing step contributes measurably.

Accuracy is below the 95–99% reported by large CNN models on the same data, but surpasses the FFT+SVM baseline (81.5%). Collapsing to a binary healthy/faulty decision gives 97.3% accuracy; only 6 faulty motors are misclassified as Healthy (5 BRB1 + 1 BRB3), concentrated at light loads where the modulation is near the noise floor.

Keywords: Motor Current Signature Analysis, broken rotor bar, LSTM, TinyML, envelope extraction, edge computing, predictive maintenance

I. Introduction

The broken rotor bar fault signature is a low-frequency amplitude modulation (2–6 Hz) of the stator current carrier (50/60 Hz). Rotor asymmetry creates sidebands at $f(1 \pm 2s)$ in the current spectrum, which manifest as a slow envelope oscillation. The envelope is therefore empirically sufficient for fault detection on this task.

The mainstream approach uses large neural networks (CNNs, ResNets, VGGs) operating on spectrograms or raw waveforms to implicitly learn this envelope extraction. We show that making the extraction explicit via a simple moving-RMS filter reduces the need for model capacity by three orders of magnitude.

A. Prior Art

Table I compares published neural-network approaches for broken rotor bar detection. The top-performing approaches (VGG-19, ResNet-152) use 60–144 M parameters to solve a 5-class signal processing problem. They achieve 98–99% accuracy but require GPU hardware and are impractical for MCU deployment. Decision trees on FFT features achieve 95–98% with fewer parameters but require explicit sideband identification, which is fragile under variable speed and noise.

Table I. *Comparison with published approaches.*

Reference	Architecture	Params	Cls	Acc	Input	Runtime
Chisedzi [1]	DT / ANN	~1–10 K	3	95–98%	FFT feat.	Desktop
Barrera [2]	VGG-19	143.7 M	5	99.4%	Spectro.	GPU
Barrera [2]	NASNet-M	5.3 M	5	96.2%	Spectro.	GPU
Jakaria [3]	CNN-LSTM	~100 K	5	92.3%	Raw I	GPU
Baseline	FFT+SVM	n/p	5	81.5%	18 FFT feat.	Desktop
Baseline	FFT+MLP	773	5	67.0%	18 FFT feat.	Desktop
This work	**LSTM h=32**	**4,773**	**5**	**88.0%**	**Envelope**	**Browser**

II. Dataset

The Broken Rotor Bar dataset, published by the Universidade Federal de Uberlândia on IEEE DataPort, contains electrical and vibration recordings from a 1 hp, 220/380 V, 4-pole, 60 Hz squirrel-cage induction motor (rated 1715 rpm, 4.1 Nm, 34 rotor bars).

Five rotor states are tested: Healthy (no defect), BRB1–BRB4 (1–4 adjacent broken bars). Each state is tested at 8 load levels (12.5–100% of rated torque) with 10 repetitions per condition. Each acquisition is 18 s long, sampled at ~55.6 kHz, covering startup transient to steady state. Only the three phase currents (Ia, Ib, Ic) are used; voltages and vibration channels are discarded.

III. Proposed Method

A. Physics of Broken Rotor Bars

Broken rotor bars introduce sidebands in the stator current at frequencies $f(1 \pm 2s)$, where f is the line frequency and s is the slip. For the test motor ($f = 60$ Hz, $s \approx 0.047$ at rated load), this gives sidebands at 54.4 Hz and 65.6 Hz. In the time domain, these manifest as amplitude modulation at $2sf \approx 5.6$ Hz. The modulation depth is proportional to k/N, where k is the number of broken bars and $N = 34$ is the total bar count: ~2.9% for BRB1, ~11.8% for BRB4. The classification task thus reduces to estimating modulation depth, which can be inferred directly from the envelope signal.

B. Envelope Extraction

The 60 Hz carrier is removed by computing the moving root-mean-square over one half-cycle (463 samples at 55.6 kHz):

$$env(t) = \sqrt{\left[(1/W) \sum x(t+i)^2 \right]}, \quad i = 0..W-1$$

For a pure sinusoid $x(t) = A \cdot \sin(\omega t)$, the half-cycle RMS equals $A/\sqrt{2}$. For an amplitude-modulated signal $x(t) = A(t) \cdot \sin(\omega t)$ where $A(t)$ varies slowly, the RMS tracks the envelope $A(t)/\sqrt{2}$.

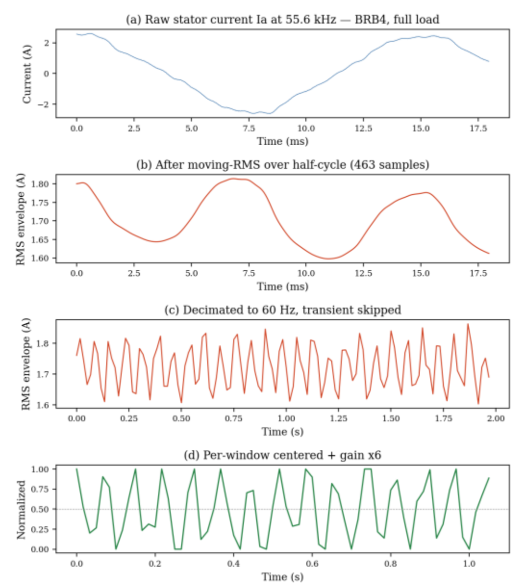

Fig. 1. Processing pipeline: (a) raw current, (b) moving-RMS envelope, (c) decimated, (d) centered.

C. Decimation and Transient Removal

The envelope is decimated by 927× to reach ~60 Hz (one sample per fundamental cycle). The first 360 samples (~6 s) are discarded to exclude the startup transient (inrush peaks of ±17 A that contaminate normalization statistics).

D. Per-Window Centering

The envelope mean encodes load level, which is a confound (all classes appear at all loads). For each 64-step window and each channel:

$$out(t) = clamp[\, (env(t) - mean(env)) \cdot G + 0.5,\ \ 0,\ \ 1\,]$$

with gain $G = 6.0$. This removes the load baseline and maps the ± 0.05 A modulation into ± 0.3 of the [0, 1] dynamic range, giving the LSTM a 10× increase in the relative magnitude of the fault signature.

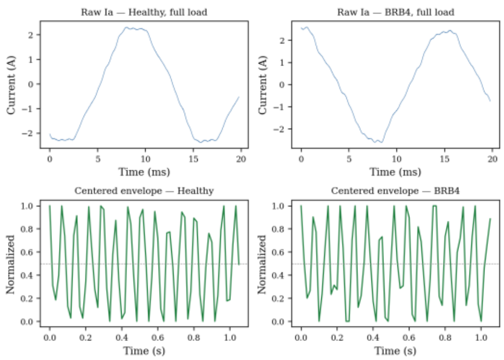

Fig. 2. Raw current (top) vs. centered envelope (bottom) for Healthy and BRB4 at full load.

E. LSTM Architecture

A single-layer LSTM with 3 inputs (Ia, Ib, Ic envelopes), 32 hidden units, and 5 sigmoid outputs. Total: 4,773 parameters (19.1 KB at float32). The model is trained with MSE loss and Adam optimizer (lr = 0.003) for 150 epochs, with one-hot targets applied at every timestep. Sigmoid + MSE was chosen for compatibility with the runtime; softmax + cross-entropy would be standard and is expected to improve results near decision boundaries.

Table II. *Parameter count breakdown.*

Component	Formula	Count
LSTM input→hidden	4 × 32 × 3	384
LSTM hidden→hidden	4 × 32 × 32	4,096
LSTM biases	4 × 32	128
Output weights	32 × 5	160
Output biases	5	5
Total		**4,773**

IV. Results

A. Ablation Study

Table III and Fig. 3 show the impact of each preprocessing step. The full envelope + centering pipeline achieves 3× the accuracy of the best raw-signal variant. Each step contributes measurably: removing any one causes a regression.

Table III. *Ablation: preprocessing impact on accuracy.*

Pipeline variant	Accuracy	Dominant failure
Raw, per-trace norm, 64-step	26.8%	Signature erased by normalization
Raw, global norm, 256-step	16.0%	Vanishing gradient (256 BPTT steps)
Envelope, global min/max	35.0%	Load confound (75% dynamic range)
Env. centered	**88.0%**	**Full pipeline**
FFT + SVM (baseline)	81.5%	Requires hand-crafted features
FFT + MLP (baseline)	67.0%	Small parametric model on FFT features

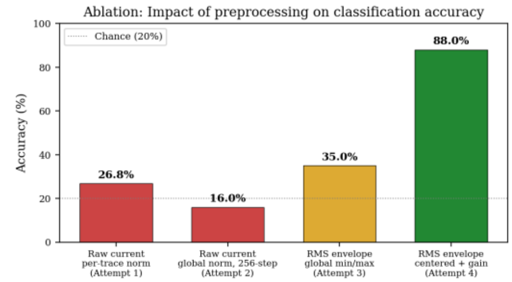

Fig. 3. Ablation: accuracy at each preprocessing stage. Dashed line = chance (20%).

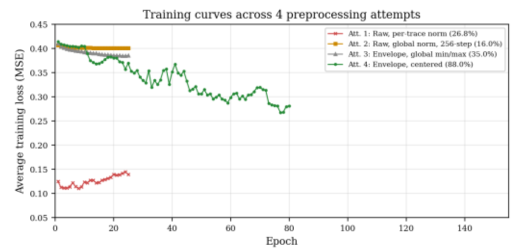

Fig. 4. Training loss across 4 attempts. Only the envelope + centering pipeline (green) converges.

B. Comparison with Classical MCSA

An FFT + SVM baseline using 18 hand-crafted spectral features (sideband amplitudes at $f(1 \pm 2s)$ for five slip values, harmonic ratios, RMS, crest factor, THD) reaches 81.5% on the same data split. The LSTM now surpasses the SVM by 6.5 percentage points, but requires explicit sideband identification and slip estimation, which depend on prior knowledge of the operating conditions. An FFT + MLP with 773 parameters (comparable to our model size) drops to 67.0%, suggesting that FFT features alone are insufficient for a small parametric model without additional structure. The LSTM learns directly from the time-domain envelope without any frequency-domain computation.

C. Classification Performance

The final model (LSTM h=32, 150 epochs) achieves 88.0% overall 5-class accuracy. Collapsing to binary healthy/faulty gives 97.3%. Healthy recall is 93.8% (75/80). BRB1 recall improved to 80.0% (72/90). Only 6 faulty motors are misclassified as Healthy (5 BRB1 + 1 BRB3), concentrated at light loads.

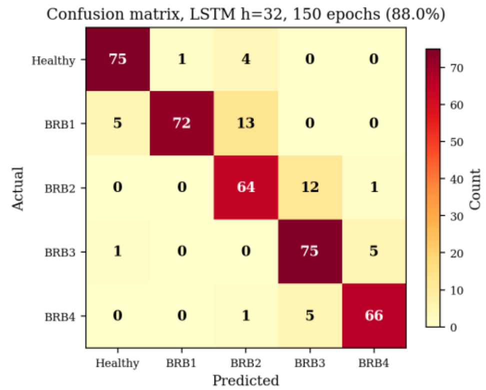

Fig. 5. Confusion matrix. Errors are concentrated between adjacent severities.

Confusions occur only between adjacent severities (BRB2↔BRB3, BRB3↔BRB4). This is the expected MCSA failure mode: the modulation difference between k and k+1 broken bars is only $1/N$ = 2.9% additional depth. BRB1 is the hardest class (80.0%), mostly confused with BRB2 at light loads where the modulation differences between adjacent severities are subtle.

D. Efficiency

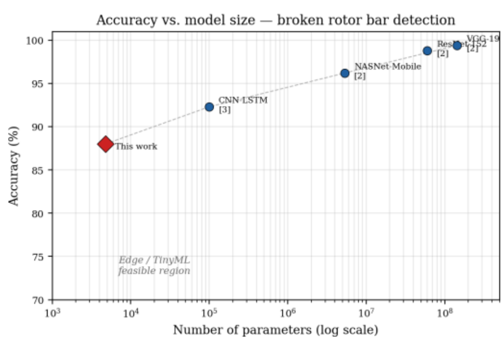

Fig. 6. Accuracy vs. model size. This work (red diamond) occupies the TinyML-feasible region.

Table IV. *Efficiency comparison.*

Model	Params	Size	Acc	Acc/10K	Inference
VGG-19 [2]	143.7 M	574 MB	99.4%	0.007	~500ms GPU
NASNet-M [2]	5.3 M	21 MB	96.2%	0.18	~50ms GPU
CNN-LSTM [3]	~100 K	400 KB	92.3%	9.2	~10ms GPU
This work	**4,773**	**19 KB**	**88.0%**	**184.4**	**1.5ms JS**

The proposed model achieves 184.4 accuracy points per 10K parameters, vs. 0.007–9.2 for published approaches. At 19.1 KB (float32), it fits in the SRAM of a Cortex-M0+ MCU.

V. Discussion

A. Preprocessing Failure Modes

The development process uncovered 7 compounding preprocessing bugs (Table V) across 5 iterative attempts. Each has a distinct physical cause and a general principle that prevents it. These failure modes are directly transferable to any time-series fault classification system.

Table V. *Documented preprocessing failure modes.*

#	Failure mode	Effect	Prevention principle
A	Per-trace normalization	Amplitude signature erased	Normalize globally when amplitude matters
B	Transient in stats	Inrush dominates range	Skip transients before computing stats
C	Window too short	Fault not visible	Window ≥ 1/f_discriminative
D	Neutral-target trap	Loss stuck at 1/C	Avoid trivial minima in loss landscape
E	Vanishing gradient	Model learns only bias	Shorten sequence via preprocessing
F	Decimation 46× off	Transient skip failed	Derive from named target rate
G	Load confound	75% range wasted	Remove nuisance variables first

B. Limitations

The 88.0% accuracy surpasses the FFT+SVM baseline (81.5%) but is below the 95–99% of large models. The gap comes from model capacity (4,773 vs. millions of parameters), suboptimal loss function (sigmoid + MSE vs. softmax + cross-entropy), and no data augmentation. Binary healthy/faulty accuracy is 97.3%. BRB1 recall improved to 80.0% (72/90); only 5 BRB1 samples remain misclassified as Healthy, concentrated at light loads.

Results are reported on a single dataset (1 motor, 34 bars, 60 Hz). Generalization requires validation on different line frequencies (50 Hz), bar counts, and variable-speed drives.

C. Browser-Native Training

The complete pipeline runs in a single HTML page using a custom JavaScript runtime: ~150 s training, 1.5 ms/sample inference, no server or GPU. This enables on-premise learning where raw currents never leave the device, interactive educational use, and rapid prototyping without ML framework installation.

VI. Deployment Context

The target platform is an ESP32-class MCU. It continuously tracks the frequency spectrum and harmonic signature of the stator current, comparing against a learned baseline. The monitoring stage does not attempt fault identification. It only detects *change*.

When a spectral shift is flagged, the device enters a diagnostic phase. Connected to a supervisory policy engine, it can receive ONNX models pushed at runtime and run them over a defined observation period. Several tiny classifiers may be loaded in sequence, each targeting a different fault family. The broken rotor bar classifier presented here is one of them. In standalone mode (no connectivity), the MCU runs whatever models are stored in flash and queues results until the link is restored.

The graph-native runtime can load and modify ONNX model topology at runtime without recompilation. This structural plasticity is what makes the architecture practical: a single MCU can host different diagnostic models depending on what the monitoring stage observed, rather than being hard-wired to one fault type.

VII. Conclusion

Extracting the stator current envelope before classification reduced the required model size from hundreds of thousands of parameters to 4,773. The preprocessing is straightforward (half-cycle moving RMS, decimation, mean subtraction) and each step has a measurable effect on accuracy, as shown in the ablation. The resulting 19 KB LSTM reaches 88.0% on 5-class severity grading and 97.3% on binary fault detection, surpassing the FFT+SVM baseline (81.5%). BRB1 at light load remains the hardest case.

On an ESP32, this classifier would run as one module in a larger pipeline: the MCU monitors harmonic signatures for spectral change, then loads the appropriate ONNX model to investigate. The 4,773 parameters fit comfortably in SRAM at any precision. For MCSA, careful signal representation can substitute for model scale.

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